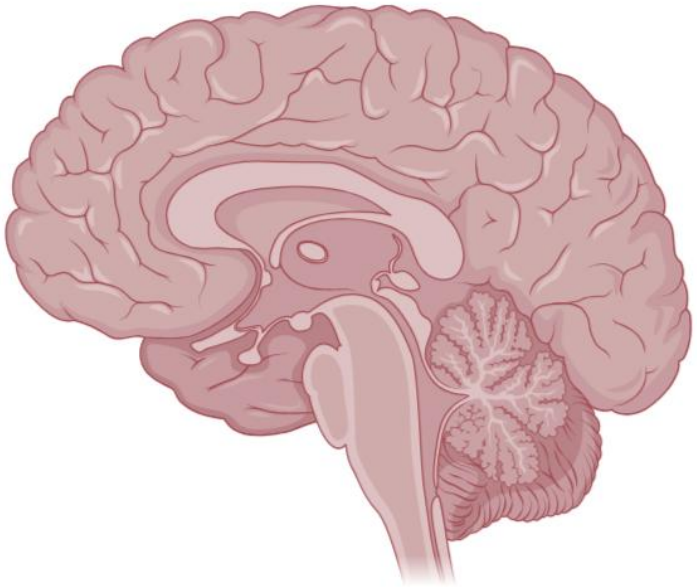


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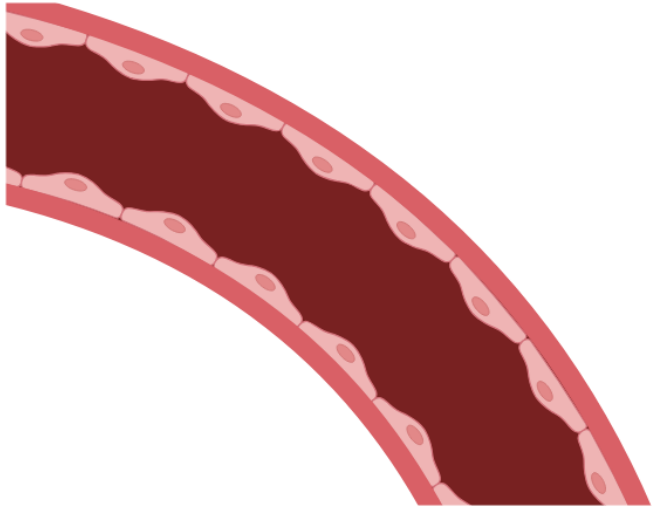
Comparative Bulk RNA Sequencing Analysis of Brain, Blood and Muscle Tissue Samples

Pogliana Giacomo

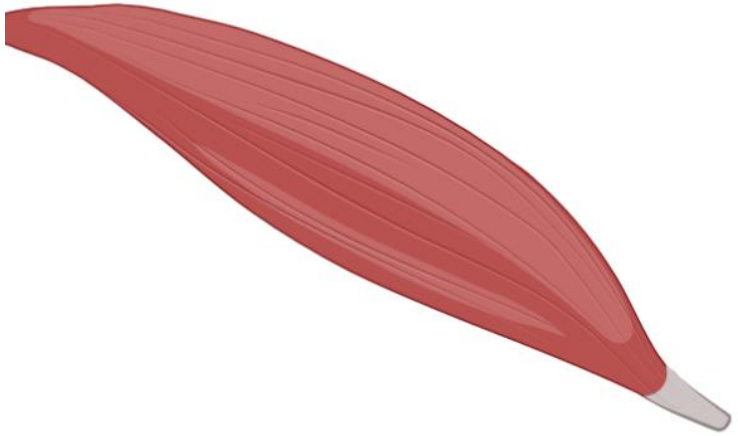
Tissue



Brain



Blood vessel



Muscle

Quality control

Parameter thresholds for sample selection:

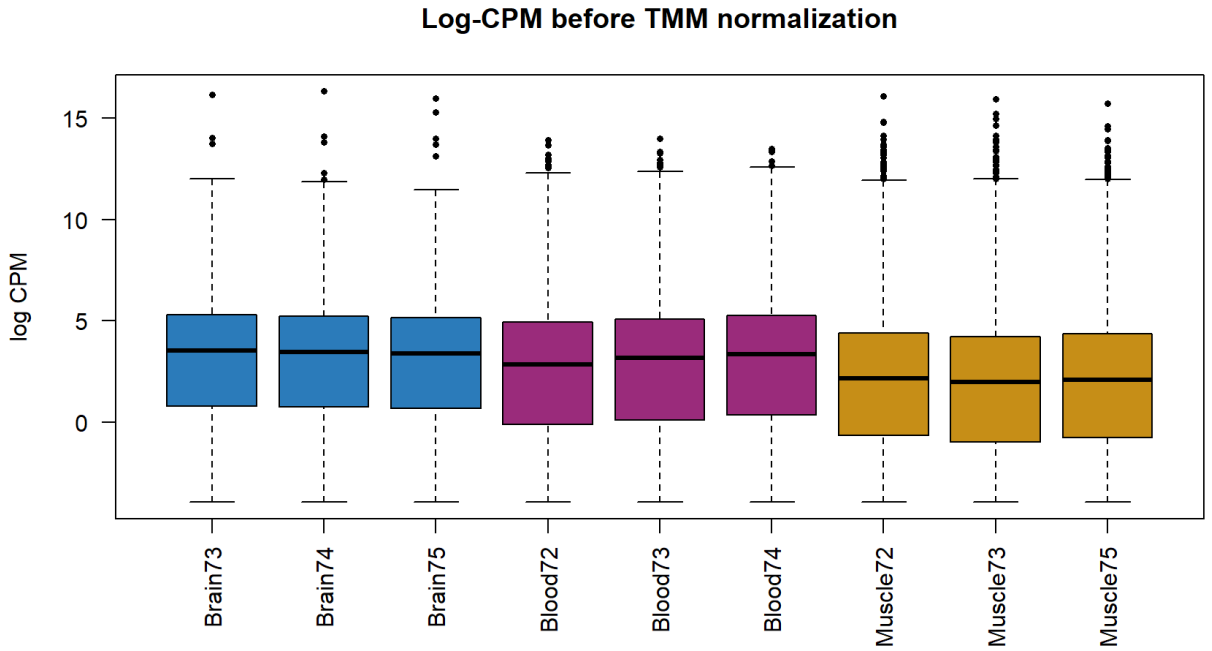
- RNA Integrity Number ≥ 6
- Mapped Reads ≥ 80 %
- Fraction of rRNA reads < 0.1

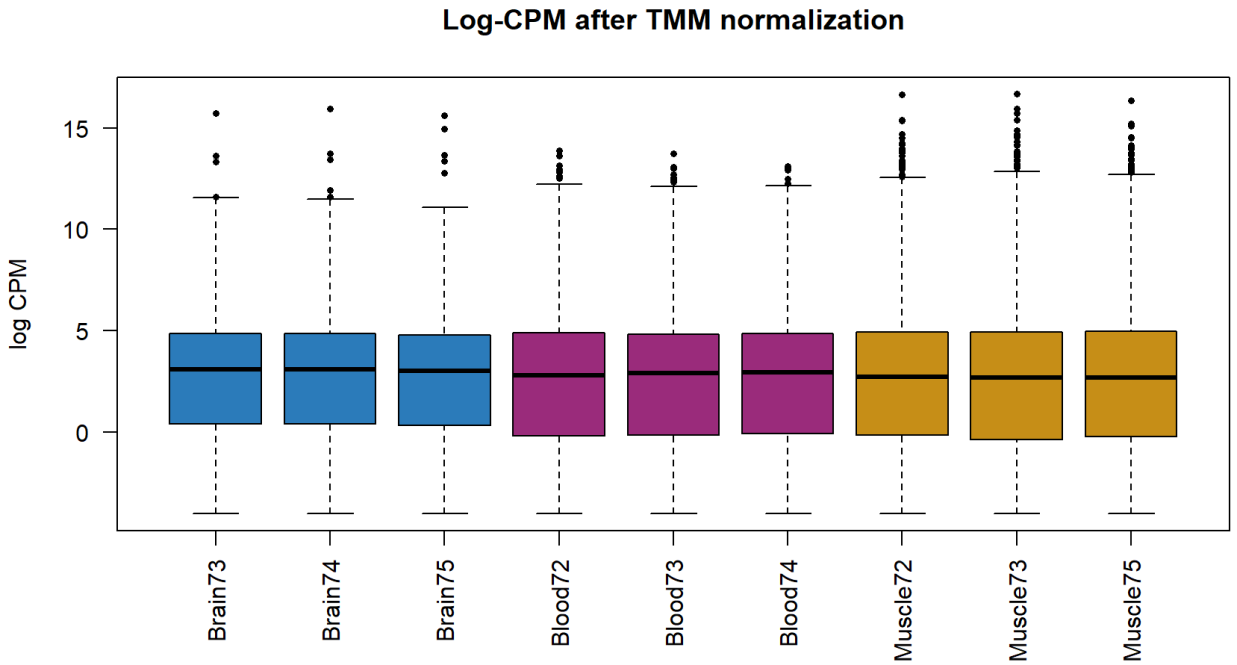
Sample	RIN	% mapped reads	% rRNA reads
Brain 72	5.9	90,3	0,11
Brain 73	6,8	88,4	0,07
Brain 74	6,8	89,1	0,07
Brain 75	6,9	91,8	0,06

Sample	RIN	% mapped reads	% rRNA reads
Blood 72	7,6	91,7	0,005
Blood 73	8,6	91,7	0,003
Blood 74	7,3	89,5	0,005

Sample	RIN	% mapped reads	% rRNA reads
Muscle 72	7,8	90,9	0,010
Muscle 73	8,5	89,8	0,010
Muscle 74	5,7	92	0,0080
Muscle 75	8,2	92,5	0,010

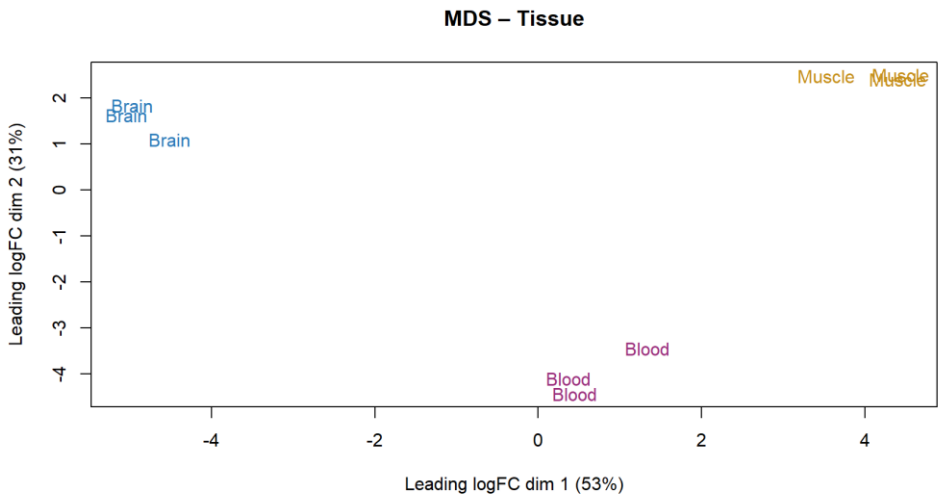
Normalization

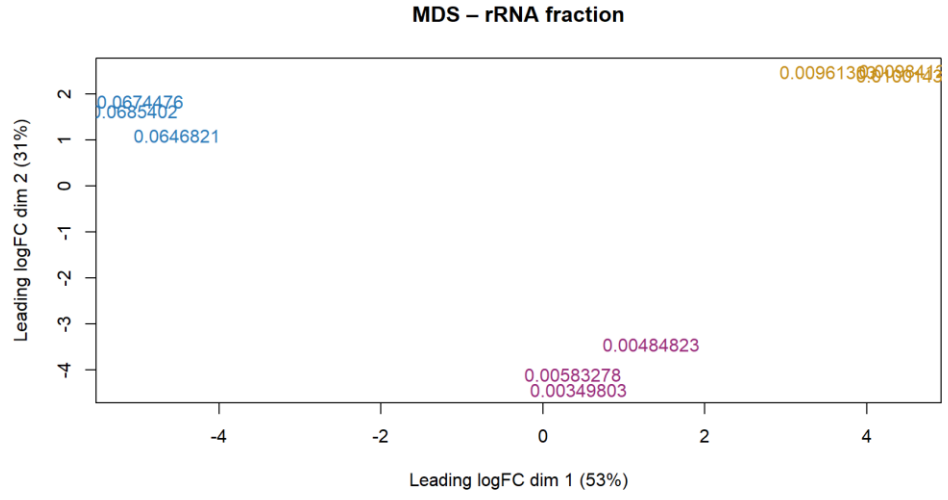


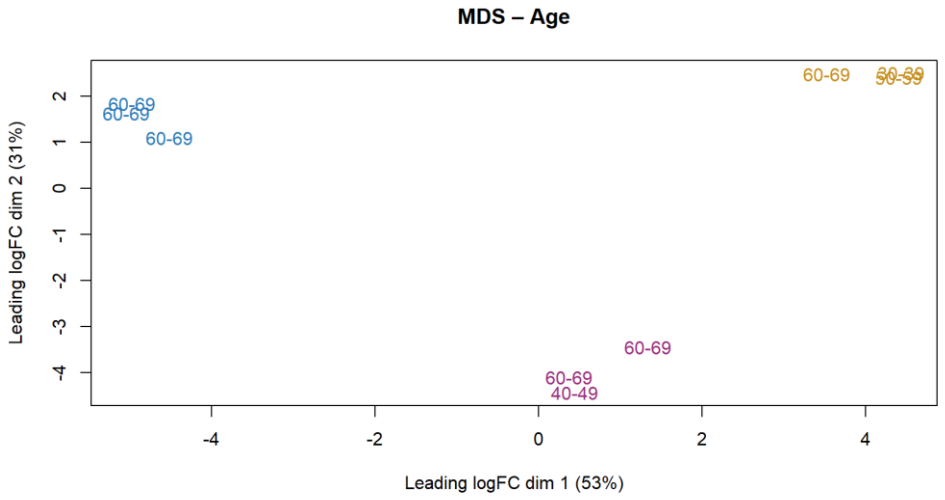


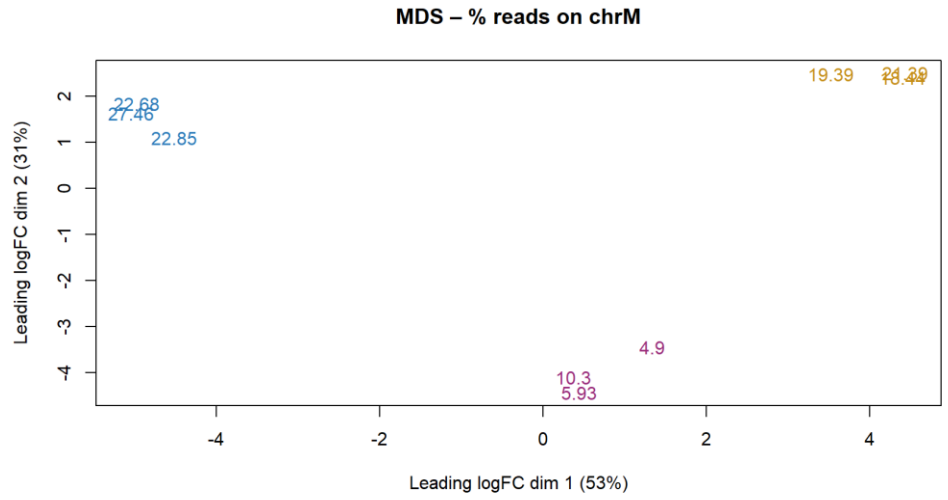
	Brain 73	Brain 74	Brain 75	Blood 72	Blood 73	Blood 74	Muscle 72	Muscle 73	Muscle 75
Normalization factors	1,3318	1,3034	1,2868	1,0353	1,1968	1,3319	0,6838	0,6046	0,6558
Library size	29.587.519	27.171.059	31.016.252	34.072.837	34.730.143	32.642.371	29.943.253	29.940.392	30.420.554
Effective library size	39.405.740	35.415.845	39.912.671	35.278.671	41.565.087	43.476.428	20.478.083	18.104.959	19.951.489

Multidimensional Scaling









Brain Blood Vessel Muscle

Differentially Expressed Genes

DE Gene definition thresholds:

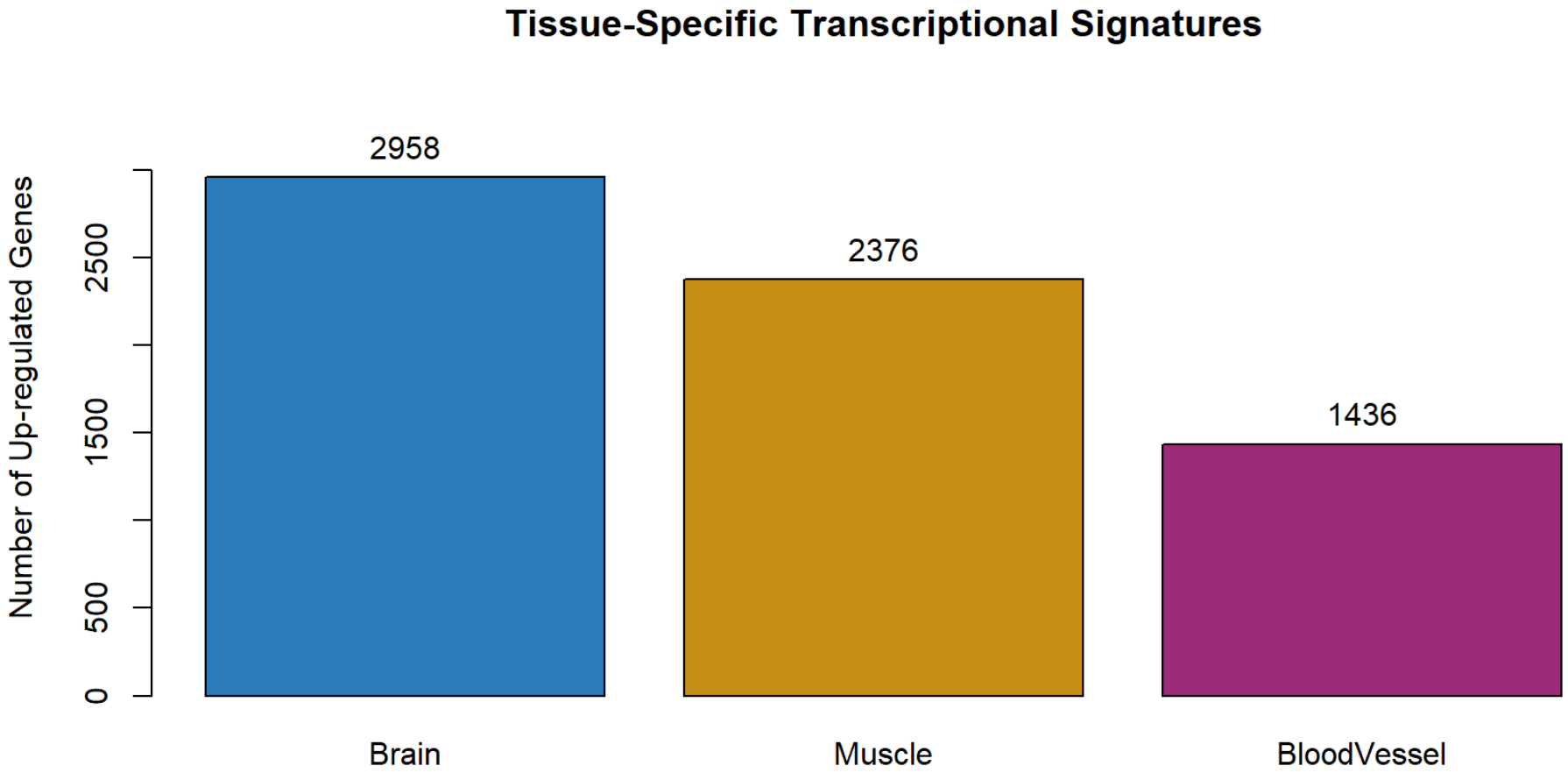
- FDR < 0.01 (Benjamini-Hochberg correction)
- |LogFC| > 1

Muscle	Brain
UP-REGULATED	DOWN-REGULATED
4199	4788

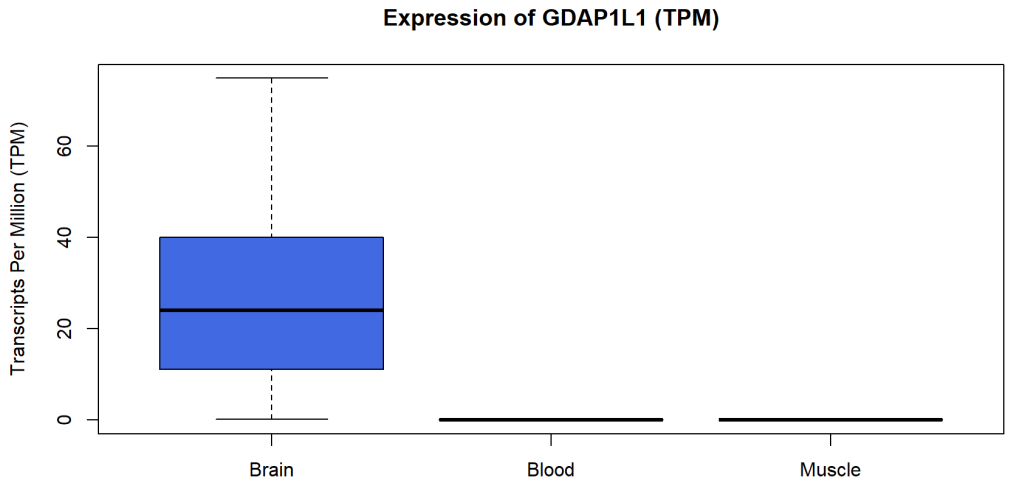
Brain	Blood vessel
UP-REGULATED	DOWN-REGULATED
4031	3157

Muscle	Blood vessel
UP-REGULATED	DOWN-REGULATED
3306	3163

Differentially Expressed Genes

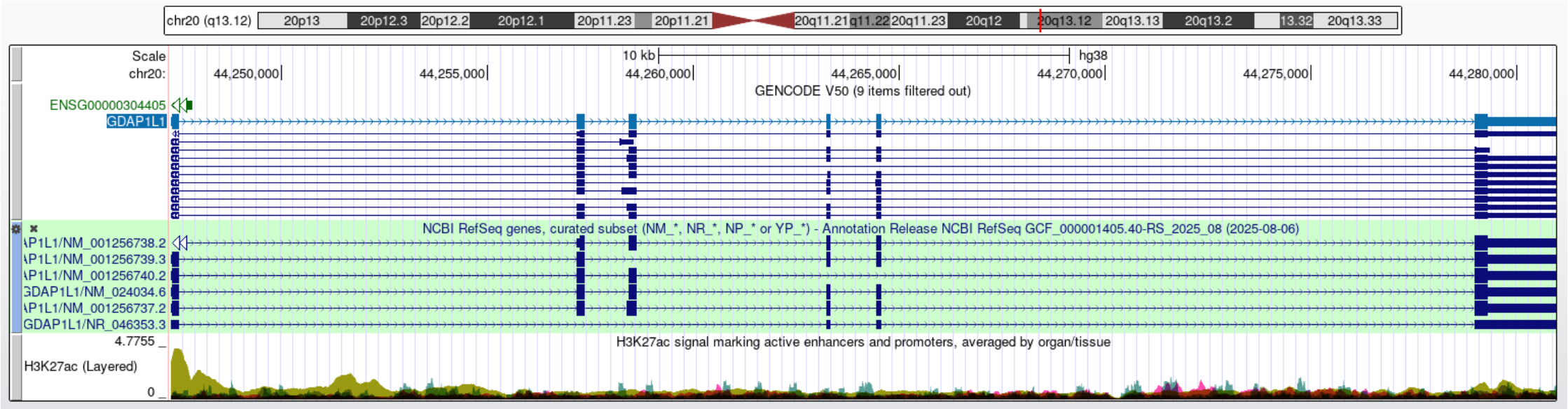


GDAP1L1 Expression



Wilcoxon test

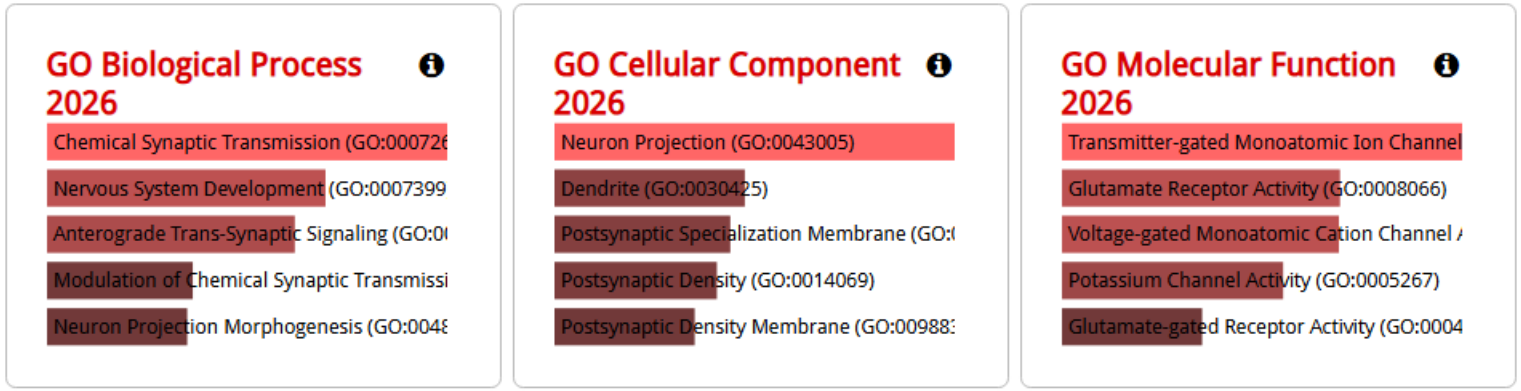
p-value < 2.2e-16



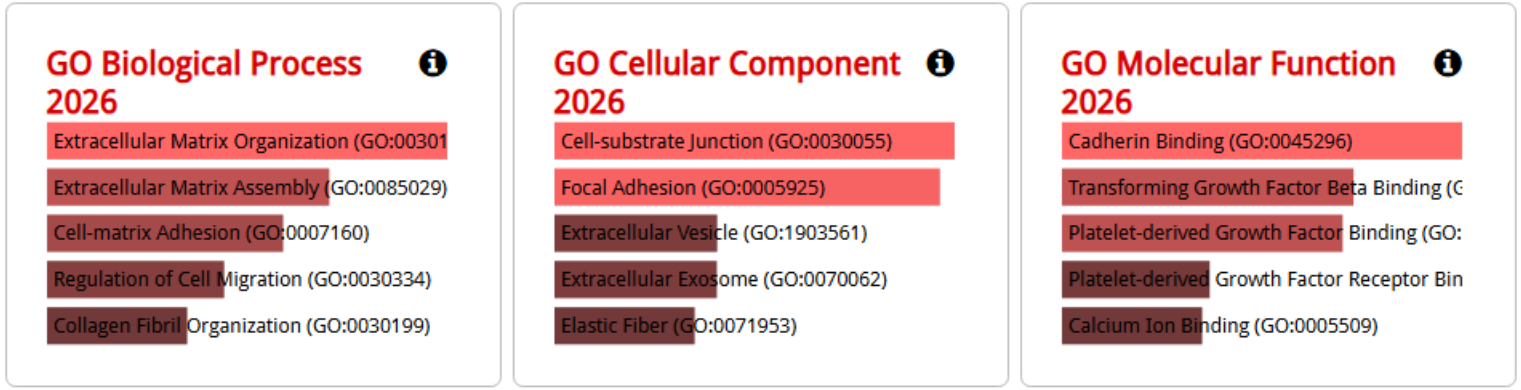
Assembly: Human Dec. 2013 (GRCh38/hg38)

Functional Enrichment Analysis on DEGs

Brain



Blood Vessel



Muscle

